

§.2.1

P.1

(a) $(4-3t)e^t$ (b) $3\sqrt{3}t \sin(\sqrt{3}t)$

(c) $2(4-3t)e^t + 12\sqrt{3}t \sin(\sqrt{3}t)$

(d) $2-3t^2$ (e) $5(2-3t^2) \neq 0$

(f) $2-3t^2$

P3. Proof: $L[cy](t) = \int_a^t s^2[cy(s)]ds$

$$= c \int_a^t s^2 y(s) ds$$

$$= cL[y](t)$$

$$L[y_1 + y_2](t) = \int_a^t s^2 [y_1(s) + y_2(s)] ds$$

$$= \int_a^t s^2 y_1(s) ds + \int_a^t s^2 y_2(s) ds$$

$$= L[y_1](t) + L[y_2](t).$$

Therefore, the operator L is linear.

P4. Proof: $L[t^2] = t+1$, $L[t] = 2t+2$.

So that

$$L[t-2t^2] = L[t] - 2L[t^2]$$

$$= (2t+2) - 2(t+1)$$

$$= 0$$

i.e.

$$(t-2t^2)'' + p(t)(t-2t^2)' + q(t)(t-2t^2) = 0$$

$\Rightarrow y(t) = t-2t^2$ is a solution of

$$y'' + p(t)y' + q(t)y = 0$$

P5. Solution:

(a) Let $L[y](t) = 2t^2 y''(t) + 3t y'(t) - y(t)$

then

$$L[\sqrt{t}] = 2t^2(\sqrt{t})'' + 3t(\sqrt{t})' - \sqrt{t}$$

$$= 2t^2(-\frac{1}{4}t^{-\frac{3}{2}}) + 3t(\frac{1}{2}t^{-\frac{1}{2}}) - t^{\frac{1}{2}}$$

$$= -\frac{1}{2}\sqrt{t} + \frac{3}{2}\sqrt{t} - \sqrt{t} = 0$$

$$L[\frac{1}{t}] = 2t^2(\frac{1}{t})'' + 3t(\frac{1}{t})' - \frac{1}{t}$$

$$= 2t^2(-\frac{2}{t^3}) + 3t(-\frac{1}{t^2}) - \frac{1}{t}$$

$$= \frac{4}{t} - \frac{3}{t} - \frac{1}{t} = 0$$

Hence, $y_1(t) = \sqrt{t}$, and $y_2(t) = \frac{1}{t}$ are the solution of $2t^2 y'' + 3t y' - y = 0$.

(b) $W[y_1, y_2](t) = y_1(t)y_2'(t) - y_1'(t)y_2(t)$

$$= \sqrt{t} \cdot (-\frac{1}{t^2}) - (\frac{1}{2\sqrt{t}}) \cdot (\frac{1}{t})$$

$$= -t^{-\frac{3}{2}} - \frac{1}{2}t^{-\frac{3}{2}}$$

$$= -\frac{3}{2}t^{-\frac{3}{2}}$$

$$\lim_{t \rightarrow 0^+} W[y_1, y_2](t) = \lim_{t \rightarrow 0^+} -\frac{3}{2}t^{-\frac{3}{2}} = -\infty$$

(c) It is clear that $p(t) = \frac{3t}{2t^2} = \frac{3}{2t}$ and

$q(t) = -\frac{1}{2t^2}$ is continuous on $0 < t < \infty$.

$$W[y_1, y_2](1) = -\frac{3}{2} \cdot 1 = -\frac{3}{2} \neq 0$$

So that $y_1(t)$ and $y_2(t)$ are linearly independent on $0 < t < \infty$. Therefore,

$y_1(t)$ and $y_2(t)$ form a fundamental set of solution of the equation on $0 < t < \infty$

(d) $2t^2 y'' + 3t y' - y = 0$, $y(1) = 2$, $y'(1) = 1$

clearly has the solution of the form.

$$y(t) = c_1 y_1(t) + c_2 y_2(t)$$

$$= c_1 \sqrt{t} + c_2 \frac{1}{t}$$

$$y(1) = 2 \Rightarrow c_1 + c_2 = 2$$

$$y'(1) = 1 \Rightarrow \frac{c_1}{2} - c_2 = 1 \Rightarrow \begin{cases} c_1 = 2 \\ c_2 = 0 \end{cases}$$

$$\Rightarrow y(t) = 2\sqrt{t}$$

P7. (a) $W[y_1, y_2](t) = -b \sin at \sin bt - a \cos at \cos bt$

(b) $W[y_1, y_2](t) = 0$

(c) $W[y_1, y_2](t) = (b-a)e^{(a+b)t}$

(d) $W[y_1, y_2](t) = e^{2at}$

(e) $W[y_1, y_2](t) = t$

(f) $W[y_1, y_2](t) = -be^{2at}$

P9. (a) Proof: $W[y_1, y_2](t) = y_1(t)y_2'(t) - y_1'(t)y_2(t)$
 $\Rightarrow W[y_1, y_2](t_0) = y_1(t_0)y_2'(t_0) - y_1'(t_0)y_2(t_0)$
 $= 1 \cdot 1 - 0 \cdot 0$
 $= 1$

Thus, $y_1(t)$ and $y_2(t)$ form a fundamental set of solutions of (3) on the interval $\alpha < t < \beta$.

(b). First, it is clear that $y(t) = y_0 y_1(t) + y_0' y_2(t)$ is a solution of (3). Now, let us verify the initial conditions.

$$y(t_0) = y_0 y_1(t_0) + y_0' y_2(t_0) = y_0$$

$$y'(t_0) = y_0 y_1'(t_0) + y_0' y_2'(t_0) = y_0'$$

Thus, $y(t) = y_0 y_1(t) + y_0' y_2(t)$ is the solution of (3) with $y(t_0) = y_0$, $y'(t_0) = y_0'$.

P10. Proof: $L[y](t) = y'' + p(t)y'(t) + q(t)y(t)$

$$\Rightarrow L[t^2] = (t^2)'' + p(t)(t^2)' + q(t)t^2$$

$$= 2 + 2t p(t) + t^2 q(t)$$

Since $p(t)$ and $q(t)$ are continuous at $t=0$, we have.

$$L[t^2](0) = 2 + 2 \cdot 0 \cdot p(0) + 0^2 \cdot q(0)$$

$$= 2 \neq 0.$$

$\Rightarrow$

$L[t^2]$ cannot be identically zero on any interval containing 0.

i.e., $y(t) = t^2$ can never be a solution of (3)